# Training AI to Understand Emotions

A Governed Multimodal Architecture for Emotional Signal Capture
and Attested Calibration in Language Model Alignment

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## Abstract

Current AI alignment techniques, including Reinforcement Learning from Human Feedback (RLHF) and Direct Preference Optimization (DPO), rely on single-scalar preference signals that conflate reasoning correctness with emotional appropriateness. A human evaluator who selects response A over response B communicates nothing about what they felt during the evaluation: whether the chosen response was merely correct or also emotionally calibrated to the interaction context. This paper argues that emotions are not innate programs to be detected by AI systems but learned associations between context, physiological state, and outcomes that must be acquired through structured multimodal feedback loops. We present a governed architecture for multimodal emotional signal capture and attested calibration comprising four signal channels (continuous emoji stream, free-text annotation, on-device voice tone analysis, and demographically gated facial expression analysis), a dual-axis training signal that separates reasoning quality from emotional salience through multiplicative weighting, a persistent Emotional Memory Cabinet with Bayesian belief updating and drift detection, and a cryptographic attestation chain that provides verifiable provenance over the complete emotional calibration trajectory. The architecture operates within a deterministic enforcement boundary that mechanically gates biometric modality activation on Fitzpatrick-stratified accuracy validation, ensuring demographic fairness is enforced architecturally rather than audited retrospectively. We describe three deployment modes spanning fully air-gapped environments to sovereign on-premises installations using Google Distributed Cloud, enabling deployment in regulated industries including healthcare, financial services, and defense. The system produces signed emotional training receipts suitable for regulatory audit, creating a governed path from emotional signal capture through model alignment that no existing system provides.

## 1. Introduction: Why AI Does Not Understand Emotions

Every major language model deployed today is emotionally blind. Not because the models lack the capacity to generate emotionally resonant text, but because the training infrastructure that shaped their behavior never captured emotional data. The human feedback that drives alignment through RLHF and DPO is a binary signal: this response is better than that response. The evaluator clicks a button. The preference is recorded. The model learns. But the infrastructure records nothing about the evaluator's emotional state during the evaluation. Was the chosen response comforting? Frustrating? Technically adequate but emotionally cold? The training pipeline does not know and cannot learn from what it does not capture.

This is not a detection problem. The AI research community has invested heavily in emotion detection: recognizing facial expressions, classifying voice prosody, analyzing text sentiment. These capabilities are mature and commercially deployed. The gap is not in recognizing that a human is upset. The gap is in using that recognition as a structured training signal that improves the model's future behavior in similar emotional contexts.

The thesis of this paper is that emotions are learned associations, not innate programs. A language model that "understands" emotions does not possess a fixed emotional ontology preprogrammed by its creators. Instead, it acquires emotional understanding the same way humans do: through repeated exposure to multimodal signals in context, with feedback that reinforces emotionally appropriate responses and attenuates emotionally inappropriate ones. The architecture presented in this paper provides the infrastructure for that learning process.

Three structural problems prevent existing alignment pipelines from supporting emotional calibration:

**Binary feedback poverty.** RLHF and DPO compress the human evaluator's experience into a single bit: chosen or rejected. This conflation makes it impossible for the training pipeline to distinguish a response that was factually wrong from a response that was factually correct but emotionally harmful. A nurse who rejects a clinical decision support response because it delivered an accurate diagnosis in a callous tone produces the same training signal as a nurse who rejects a response because the diagnosis was incorrect. The model cannot learn the difference.

**Single-modality limitation.** Where emotional signals are captured in existing systems, they rely on a single channel, typically text sentiment analysis. Human emotional expression is inherently multimodal. Voice tone, facial expression, explicit reactions, and contextual annotations carry complementary information. A patient who says "I'm fine" while their voice tightens and they tap an angry emoji is communicating something that no single-channel system can capture.

**Absence of governed provenance.** No existing system provides cryptographic attestation over emotional training signals. In regulated industries where AI decision-making is subject to audit, there is no verifiable chain of evidence showing what emotional feedback was collected, from whom, under what consent conditions, with what demographic accuracy validation, and how that feedback influenced model behavior over time.

This paper presents an architecture that addresses all three problems simultaneously. The architecture captures emotional signals through four multimodal channels, separates reasoning quality from emotional salience as independent training dimensions with multiplicative weighting, and wraps every step in a cryptographic attestation chain that produces auditable receipts suitable for regulatory compliance in healthcare, financial services, and defense.

## 2. Related Work

### 2.1 Alignment Methods and Their Emotional Limitations

Reinforcement Learning from Human Feedback (RLHF) [1] trains a reward model on pairwise human preferences and uses it to fine-tune a language model via proximal policy optimization. Direct Preference Optimization (DPO) [2] eliminates the reward model by directly optimizing the language model on preference pairs. Kahneman-Tversky Optimization (KTO) [3] further simplifies to unpaired binary feedback. All three methods treat the human evaluator as a source of preference rankings. None capture what the evaluator felt during the evaluation.

Recent work on multi-objective alignment (ArmoRM [4], PERM [5]) decomposes the reward signal into multiple dimensions, but these dimensions are defined by the system designer (helpfulness, harmlessness, honesty) rather than derived from the evaluator's emotional experience. The emotional dimension remains absent from the objective function.

### 2.2 Emotion in AI Systems

Emotion detection is a mature field with commercial deployments. Affectiva (now Smart Eye) demonstrated multimodal CNN architectures for simultaneous audio-video emotion classification [6]. Hume AI built emotion data collection infrastructure with demographic diversity tagging [7]. Blueskeye AI fused facial, voice, and text sentiment signals into valence-arousal-dominance representations for LLM prompt customization at inference time [8].

These systems share a common limitation: they use detected emotions to modify system behavior at inference time (adjusting tone, selecting responses) rather than feeding emotional signals back into the training pipeline to improve the model's fundamental emotional understanding. Detection without training feedback produces systems that react to emotions without learning from them.

### 2.3 Emotion as a Training Signal

Sony Interactive Entertainment disclosed a system that uses detected human emotions during RLHF evaluation to discount inconsistent evaluator feedback [9]. When a human evaluator's detected emotion conflicts with their rating input beyond an inconsistency threshold, the

evaluator's selection is discounted or discarded. This system treats emotion detection as a credibility filter on human labelers, not as a training signal in its own right. The emotional state is noise to be filtered, not signal to be learned.

Google disclosed obtaining implicit feedback signals during sequential AI-generated content presentation and using them to determine whether to include a training example [10]. The feedback signals are binary (include or exclude) and are not constrained to an emotional vocabulary. Iren and Aysolmaz disclosed using detected changes in emotional state as reinforcement learning feedback, marking interaction triples as unsatisfactory when negative emotional change is detected [11]. This system is limited to binary positive-negative emotion changes with no multimodal fusion, no persistent profiles, and no provenance chain.

Meta's RLUF (Reinforcement Learning from User Feedback) [12] incorporates emoji reactions as training signals, but uses them as scalar rewards rather than as one channel in a multimodal emotional capture system. None of these approaches provide the combination of multimodal capture, governed biometric gating, dual-axis training signal separation, persistent emotional memory with Bayesian updating, and cryptographic attestation that the present work introduces.

### 2.4 Autonomous Research Loops

Karpathy's autoresearch framework [13] demonstrates autonomous AI-driven optimization of ML training configurations, achieving measurable efficiency gains through iterative agent-driven experimentation. The pattern of autonomous hypothesis generation, execution, evaluation, and iteration directly parallels the closed-loop emotional calibration cycle described in Section 5. The critical difference is governance: autoresearch operates without attestation, audit trails, or demographic fairness enforcement. The architecture presented here applies the autonomous optimization pattern within a governed framework suitable for regulated deployment.

### 2.5 Cryptographic Training Data Provenance

The landscape for cryptographic attestation of training data is actively developing. Russinovich and Kelly disclosed signed training data generation via a signing authority and ledger [14]. Intel disclosed provenance-derived feature filtering to defend against data poisoning

[15]. Dartmouth researchers combined trusted execution environment attestations with blockchain for health data provenance [16]. None of these apply cryptographic attestation to biometrically derived emotional signals specifically, and none create per-event attestation topologies binding consent receipts, demographic validation receipts, signal capture receipts, and training weight update receipts into a single verifiable chain.

## 3. System Architecture

The architecture operates in two layers. Layer 1 runs on-device in real time, capturing emotional signals through four channels and processing them locally without transmitting raw biometric data. Layer 2 runs post-attestation, embedding the structured receipt content through semantic embedding models for retrieval indexing and training pipeline consumption. The two layers are connected by the emotional training receipt, a cryptographically signed artifact that crosses the boundary between on-device privacy and cloud-based retrieval.

### 3.1 Four-Channel Multimodal Emotional Signal Capture

The system captures emotional signals through four channels organized into active channels requiring explicit user action and passive channels capturing emotional cues the user emits naturally during interaction.

**Channel 1: Continuous Emoji Stream (Active)**

The first channel captures a timestamped sequence of discrete emoji signals from a controlled vocabulary. Unlike point-in-time feedback systems that collect a single reaction per interaction, this channel produces a continuous stream. The user may signal emotional responses at any point during the interaction. The system prompts the user for an emoji signal at key decision points, defined as moments where the model output may have emotional significance: delivery of a clinical result, communication of an approval or denial, presentation of a risk assessment.

The controlled vocabulary comprises a universal base palette of six emotional categories (heart, positive acknowledgment, neutral, negative acknowledgment, strong negative, and urgency alert), each mapped to a continuous sentiment weight on a scale from negative one to positive one. Domain-specific extension palettes add emoji signals tuned to particular regulated verticals. For healthcare: compassion, concern, and gratitude. For security operations: urgency, protective, and anxiety. For financial services: frustration, resolution, and loss.

**Channel 2: Free-Text Annotation (Active)**

The second channel captures optional free-text annotations linked to specific emoji signals in the stream. Text annotations carry emotional information that discrete emojis alone cannot

express, including mixed emotions, contextual reasoning, sarcasm, resignation, and frustration at process rather than outcome. Each annotation is analyzed for overall sentiment score, detected emotion categories from a multi-label taxonomy, and sarcasm probability.

**Channel 3: Voice Tone Analysis (Passive, On-Device)**

The third channel captures emotional signals from the user's voice during spoken interactions. A speech emotion recognition model (emotion2vec [17] or Wav2Small [18]) runs entirely on the user's device, producing emotional embedding vectors. Raw audio never leaves the device. Only derived emotional vectors, which cannot reconstruct the original audio or identify the speaker, are retained.

The on-device processor performs two computations. First, a windowed summary: a rolling weighted average of emotional vectors over a configurable time window (default thirty seconds), producing a baseline emotional tone. Second, spike detection: when the magnitude delta between a new vector and the rolling mean exceeds a configurable threshold (default two standard deviations), the vector is preserved as a discrete spike event with a timestamp, the full emotion vector, the delta from baseline, and a text context window. Spike events represent moments of sharp emotional shift: a voice tightening during adverse clinical information, a tone breaking during a denial communication. These events carry disproportionate emotional training signal relative to baseline summaries.

**Channel 4: Facial Expression Analysis (Passive, Gated)**

The fourth channel captures emotional signals from the user's facial expressions. A facial expression recognition model runs entirely on the user's device. Raw image frames never leave the device. This channel is distinguished from all other channels by a mandatory activation gate described in Section 3.2. The gate mechanically refuses any signal production from this channel unless explicit user consent and validated demographic accuracy are cryptographically attested.

### 3.2 Fitzpatrick-Stratified Demographic Accuracy Validation Gate

The Demographic Accuracy Validation Gate is a deterministic enforcement boundary that mechanically refuses activation of the facial expression analysis channel unless two cryptographically attested receipts are verified. This gate implements the enforcement pattern disclosed in our prior work on deterministic execution-bound governance [19].

The first required receipt is a Consent Receipt proving the user explicitly opted into facial expression analysis for the current interaction context. The second required receipt is a Fitzpatrick Stratified Validation Receipt proving the facial expression recognition classifier has been evaluated on the deployment population and meets a configurable accuracy threshold.

The validation process computes the absolute accuracy delta between benchmark performance on Fitzpatrick I-II populations and deployment performance on the target population. If the delta exceeds a configurable maximum (exemplary default: five percentage points) or the validation sample size falls below a configurable minimum (exemplary default: two hundred samples), the gate mechanically refuses activation. The refusal is itself an attested receipt in the provenance chain.

This mechanism transforms facial expression analysis from an uncontrolled liability into a governance-gated modality with cryptographic proof of validated fairness. A regulator or auditor may verify when the validation was performed, on what data, with what accuracy results, and whether the gate passed or refused. The gate is enforced at the type system level in the implementation: the FacialSignal type cannot be constructed without a validated gate pass token, making ungated facial signal capture a compile-time error rather than a runtime check.

### 3.3 Two-Layer Embedding Architecture

The architecture employs a two-layer embedding strategy that balances on-device privacy with cloud-scale semantic retrieval.

#### Layer 1: EmbeddingGemma (Local, Air-Gapped)

For the base deployment, all text embedding uses EmbeddingGemma, an open-weight, sub-500M parameter model that supports Matryoshka Representation Learning (MRL) with truncation to 128 or 256 dimensions. EmbeddingGemma runs locally with zero network

dependency, preserving the "data never leaves" guarantee for air-gapped deployments. Compact 128-256 dimensional vectors are stored directly in the Merkle-chained receipt for fast deterministic lookup.

**Layer 2: Gemini Embedding 2 (Sovereign On-Premises)**

For enterprise sovereign deployments, Gemini Embedding 2 [20] runs on Google Distributed Cloud (GDC) hardware located entirely within the client's data center with no connection to the public internet. Gemini Embedding 2 is a natively multimodal embedding model that maps text, images, video, audio, and documents into a single 3072-dimensional vector space. The full-resolution vectors are stored as linked artifacts for the training pipeline and deep semantic search.

Three deployment modes are supported: AirGapped (EmbeddingGemma only, no cloud models), GoogleDistributedCloud (Gemini Embedding 2 on-premises, no public internet), and CloudApi (public Gemini API for development and non-regulated deployments only). MRL tiered receipts carry compact vectors in the receipt chain and full-resolution vectors for the training pipeline, ensuring the receipt chain remains lightweight while the training infrastructure receives maximum signal.

## 4. Dual-Axis Training Signal Architecture

The central innovation in the training signal design is the separation of model evaluation into two independent axes, each computed from different signal sources and each optimizing a different aspect of model behavior.

### 4.1 Score A: Reasoning Quality

Score A measures objective execution quality and is computed deterministically by the governance kernel without human input. The score is derived from whether the model's output was valid according to the operational schema, whether execution succeeded, whether a human operator corrected the output, and the efficiency of the output relative to population percentiles.

Score A produces a discrete label on a four-tier scale: Gold (valid, successful, no correction, above-median efficiency), Silver (valid, successful, minor correction or below-median efficiency), Bronze (valid, successful, significant human correction), and Miss (schema invalidity, execution failure, or governance refusal). These tiers map to normalized weights: Gold = 1.0, Silver = 0.7, Bronze = 0.4, Miss = 0.0.

### 4.2 Score B: Emotional Salience

Score B measures emotional appropriateness and is derived from the multimodal emotional signals captured through the four-channel system. The fused emotional valence, computed as a weighted aggregate across all active channels, produces a continuous score from negative one to positive one. This score is normalized to a weight between zero and one using the mapping: weight = (valence + 1.0) / 2.0.

### 4.3 Multiplicative Composition

The dual-axis architecture combines Score A and Score B through multiplicative weighting:

```
CompositeWeight = NormalizedScoreA x NormalizedScoreB
```

This multiplicative composition creates a training signal that requires both reasoning correctness and emotional calibration for maximum weight. An interaction that is technically correct (high Score A) but emotionally inappropriate (low Score B) receives reduced weight. An interaction that is emotionally appropriate (high Score B) but technically flawed (low Score A)

also receives reduced weight. Only interactions that score highly on both axes receive maximum DPO chosen weight.

The multiplicative structure is distinct from additive multi-objective reward composition used in prior alignment systems [4]. Additive composition allows a model to compensate for low emotional quality with high reasoning quality. Multiplicative composition does not permit such compensation: a zero on either axis produces a zero composite. This property ensures that neither dimension can be ignored during training.

# 5. Emotional Memory Cabinet and the Closed Calibration Loop

## 5.1 Persistent Per-Entity Emotional Profiles

The Emotional Memory Cabinet is a persistent, per-entity emotional profile store that maintains sentiment-weighted interaction histories. Entities may include patients, locations, cases, devices, operators, or any domain-specific object that participates in recurring interactions with the AI system.

Each entity's emotional profile comprises a cumulative sentiment score, a sentiment trend indicator (declining, stable, or improving) computed over a configurable window, an array of high-weight emotional memories with associated receipt identifiers and temporal decay factors, interaction count, and timestamps of the last positive and negative events. The temporal decay factor implements exponential decay, ensuring recent emotional signals carry more weight than historical ones while preserving long-term emotional trajectory information.

## 5.2 Bayesian Belief Updating

The Cabinet is updated via Bayesian belief updating after each interaction:

```
Posterior(entity) = Prior(entity) + LearningRate x (Evidence - Prior)
                                    x Weight
```

where Prior is the entity's existing cumulative sentiment, Evidence is the aggregate emotional valence from the current interaction's signal bundle, and Weight is a multiplier that incorporates spike weighting. When voice spike events are present (indicating sharp emotional shifts), the weight is multiplied by a configurable spike multiplier (default 2.5x), reflecting the established finding in affective neuroscience that emotionally salient events are preferentially consolidated in memory.

## 5.3 The Closed Loop

The Emotional Memory Cabinet creates a closed loop with the model inference pipeline. At inference time, the Cabinet's emotional profile for the relevant entity is injected into the language model's prompt, providing emotional context that influences the model's response. The model's response generates a new interaction. The new interaction produces new multimodal emotional

signals from the user. Those signals are fused and used to update the Cabinet. The updated Cabinet influences the next inference.

Every step in this loop produces or consumes a cryptographically signed receipt, creating a complete, auditable history of how the model's emotional understanding of any given entity evolved over time, through what evidence, and with what outcomes. This is not a theoretical property: the receipt chain is implemented as an append-only Merkle-anchored ledger where each receipt carries the hash of the previous receipt, enabling independent verification of chain integrity.

### 5.4 Drift Detection and Attested Recalibration

The system monitors alignment between the Cabinet's predictions and incoming emotional signals. At each interaction, the system computes the absolute divergence between the predicted emotional valence (from the Cabinet) and the actual emotional signal received. A running cumulative divergence is maintained over a sliding window of N consecutive interactions (default N = 10).

When cumulative divergence exceeds a configurable threshold (default 0.30), a Cabinet Recalibration Event is triggered. The recalibration adjusts the entity's emotional profile toward the actual signal trajectory and generates an attested receipt recording the profile before recalibration, the profile after recalibration, the divergence trajectory, and the signal evidence that motivated the correction.

This mechanism provides two capabilities critical for regulated deployment. First, it ensures the model's emotional calibration self-corrects rather than accumulating bias. Second, it produces an auditable artifact for regulators demonstrating that the system actively monitors and corrects its own emotional assessments.

### 5.5 Connection to Autonomous Research Loops

The closed calibration loop described above parallels the autonomous research pattern demonstrated by Karpathy's autoresearch framework [13]. In both architectures, an AI system iteratively modifies its behavior, evaluates outcomes against a measurable criterion, retains improvements, and discards regressions. The critical distinction is governance: autoresearch

optimizes a training script against validation loss with no attestation or audit trail. The present architecture optimizes emotional calibration against the dual-axis composite score, with every iteration producing a signed receipt that chains into a verifiable provenance ledger.

The practical implication is that the emotional calibration loop can run autonomously: new interactions produce emotional signals, the signals update the Cabinet, the updated Cabinet produces better emotional context for the next interaction, and the drift detection mechanism self-corrects when calibration degrades. This creates a governed, attested, self-improving emotional calibration system that can operate without human intervention while maintaining full auditability.

# 6. Deployment in Regulated Industries

## 6.1 Healthcare

In healthcare, an AI system that provides clinically correct information in an emotionally inappropriate manner may cause patient distress, reduce treatment adherence, or erode trust in care providers. The dual-axis training signal addresses this directly: a clinical decision support response that is medically accurate (Gold reasoning) but delivered without recognizing a patient's documented anxiety history (low emotional salience) receives reduced training weight. Over time, the model learns that clinical accuracy alone is insufficient.

The Fitzpatrick validation gate is particularly relevant in healthcare, where patient populations may span all six Fitzpatrick skin type categories. The gate ensures that facial emotion recognition used in any deployment has been validated for the specific population being served, with cryptographic proof of the validation available for HIPAA audit.

## 6.2 Financial Services

Insurance claim denials, mortgage processing decisions, and fraud investigation interviews all involve humans who have emotional responses to outcomes. A model that recognizes frustration in a claims interaction and adjusts its communication approach, with an auditable trail of how it learned that behavior, is a compliance asset. The Consumer Financial Protection Bureau's position that AI systems must comply with consumer protection laws makes emotional calibration directly relevant to regulatory risk.

The receipt chain provides the audit infrastructure that financial regulators require. A compliance officer can trace the complete emotional calibration trajectory for any entity: what emotional feedback was collected, how the model's emotional profile for that entity evolved, when drift was detected, and what recalibration actions were taken.

## 6.3 Defense and Security Operations

In security operations, operator stress detection represents a safety capability. A drone operator, a cyber defense analyst, or a logistics coordinator under pressure benefits from a system that detects voice stress patterns and adjusts its communication style. The attested receipt

chain means the after-action review can verify whether the system recognized operator stress and how it responded.

The air-gapped deployment mode (EmbeddingGemma local, no cloud dependency) is mandatory for classified environments. The entire emotional signal capture, processing, Bayesian updating, and drift detection pipeline runs on-premises with zero network transmission of any data, biometric or otherwise.

### 6.4 EU AI Act Compliance

Article 5(1)(f) of the EU AI Act prohibits AI systems that infer emotions in workplace and educational settings, with exceptions for medical and safety purposes. The architecture is designed with this regulation in mind. The emotional signal capture system does not infer emotions from biometric data in prohibited contexts. It captures explicit user feedback (emoji stream, text annotation) as active signals and processes biometric data (voice, facial) only with gated consent, validated demographic accuracy, and on-device processing that never transmits raw biometric data. The Fitzpatrick gate and consent receipt chain provide the documentary evidence required to demonstrate compliance with Article 5 exceptions for medical and safety applications.

## 7. Proposed Experimental Evaluation

This section describes a proposed evaluation methodology. The architecture has been implemented in Rust (95 tests passing across five library crates) but has not yet been evaluated on live interaction data. The experimental design is presented to enable reproducibility and to invite collaboration from research groups with access to suitable interaction datasets.

### 7.1 Proposed Metrics

Three metrics are proposed for evaluating the emotional calibration system:

**Emotional Calibration Error (ECE).** The mean absolute difference between the model's predicted emotional appropriateness score and human evaluator ratings of emotional appropriateness, measured across a held-out test set of interactions. Lower ECE indicates better emotional calibration.

**Dual-Axis Alignment Gain.** The improvement in composite DPO chosen weight (reasoning quality times emotional salience) after training with multimodal emotional signals versus training with standard binary RLHF feedback alone. This measures whether the dual-axis signal produces models that are both more correct and more emotionally appropriate.

**Drift Recovery Time.** The number of interactions required for the drift detection and recalibration mechanism to restore emotional calibration after an intentional perturbation (simulating a population shift or context change). Faster recovery indicates more robust self-correction.

### 7.2 Proposed Experimental Conditions

The evaluation compares four conditions: (1) baseline RLHF with binary preference feedback, (2) RLHF with emoji-only emotional signals (single-channel active feedback), (3) RLHF with full four-channel multimodal emotional signals and dual-axis scoring, and (4) condition 3 with the Emotional Memory Cabinet and closed-loop calibration active. Comparison of conditions 1 through 4 isolates the contribution of each architectural component.

### 7.3 Proposed Datasets

Suitable evaluation domains include clinical decision support interactions (healthcare), insurance claim processing interactions (financial services), and security operations center communications (defense). Each domain provides naturally emotional interaction contexts with clear criteria for both reasoning correctness and emotional appropriateness. Anonymized interaction data from ExecLayer's QueueFlow Sentinel platform, currently deployed in partnership with Amazon Ring for security operations monitoring, is one candidate source.

## 8. Conclusion

The training infrastructure problem, not the detection problem, is what holds back emotionally calibrated AI. Mature emotion detection technologies exist across voice, face, and text modalities. What does not exist is a governed pipeline that captures those signals as structured training data, separates reasoning quality from emotional salience as independent training dimensions, maintains persistent emotional memory with self-correcting calibration, and wraps every step in cryptographic attestation suitable for regulated deployment.

This paper presented that pipeline. The architecture introduces four novel contributions: (1) a four-channel multimodal emotional signal capture system with a Fitzpatrick-stratified demographic accuracy validation gate that enforces fairness architecturally rather than auditing it retrospectively, (2) a dual-axis training signal with multiplicative composition that requires both reasoning correctness and emotional calibration for maximum training weight, (3) a persistent Emotional Memory Cabinet with Bayesian belief updating and drift detection that creates a self-correcting closed loop between emotional signal capture and model inference, and (4) a cryptographic attestation chain that produces verifiable provenance over the complete emotional calibration trajectory.

The architecture has been implemented in Rust as five library crates with 95 passing tests across four adversarial test categories. Three deployment modes support environments ranging from fully air-gapped to sovereign on-premises installations using Google Distributed Cloud. The system is designed for integration with existing deterministic enforcement kernels, providing the emotional calibration dimension that current governance architectures lack.

Emotions are learned associations. An AI system that understands emotions must learn them through structured, multimodal, governed feedback loops, not through detection alone and not through unattested training data. The architecture presented here provides the infrastructure for that learning. The receipt chain provides the proof.

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